

Literature Review

1

Title: *Enhancing building product inventory automation through a hybrid spatial-semantic modeling framework*

Journal: *Developments in the Built Environment* (2026)

Main Goal: Finding a way to automatically scan, identify, and map building materials (like doors, windows, and panels) so they can be reused or recycled instead of thrown away when a building is renovated or demolished.

Project Objectives

- **Automate Building Inventories:** Automatically convert raw 3D room scans into structured inventory lists for building components (like doors, windows, and panels).
- **Bridge Spatial and Semantic Data:** Combine 3D spatial location coordinates (where an item is physically located and its size) with detailed semantic information (what the item is, what material it's made of, and its current condition).
- **Support Circular Construction:** Provide construction managers with fast, accurate data to evaluate which building parts can be salvaged, recycled, or reused before renovation or demolition.

Research Gap

- **Incomplete Single-Method Solutions:** Existing automated inspection tools typically address only one side of the problem:
 - **3D Scanning / AI Detection Tools** can pinpoint *where* an object is in space, but fail at capturing fine details, material properties, or physical condition.
 - **Vision-Language Models (VLMs)** excel at describing *what* an object is and reading image details, but lack 3D spatial awareness to determine exact physical locations and sizes.
- **Disconnect Between Geometry and Information:** Prior approaches lack a unified framework that seamlessly merges precise 3D geometric mapping with deep, multi-modal semantic understanding.

Research Questions

- **1:** How can 3D point cloud spatial data be reliably linked with 2D photo semantic details without losing position or dimensional accuracy?
- **2:** How accurately can an automated hybrid framework identify reusable building components compared to traditional human-led inspections?
- **3:** Does combining spatial mapping with semantic AI significantly reduce the time, cost, and effort required to create material passports for existing buildings?

My questions -

2

Title:An End-to-End Object Detection System in Indoor Environments Using Lightweight Neural Network

Main Goal

- To develop a lightweight, real-time indoor object detection system and a tailored 25-class dataset to assist visually impaired and blind individuals (VIP) in navigating indoor spaces safely and independently.

Project Objectives

- **Develop a Lightweight Object Detection Framework:** Design an end-to-end deep convolutional neural network based on a modified YOLOv3 architecture—replacing the heavy Darknet-53 backbone with MobileNet V1—to significantly reduce computational complexity, parameter count, and power consumption for resource-constrained devices.
- **Construct a Specialized Indoor Dataset:** Build and annotate a new, comprehensive dataset comprising 11,000 images covering 25 crucial indoor landmark object categories, specifically curated under varied lighting conditions, complex backgrounds, severe occlusions, and diverse viewpoints to support assistive navigation.
- **Enhance Assistive Indoor Navigation:** Provide blind, visually impaired, and sighted individuals with real-time (up to 59 FPS), highly accurate (89.78% mAP) spatial and category information about indoor obstacles, landmark features, and hazardous scenarios (such as stairways or hallway obstructions) to facilitate safer and autonomous navigation.

Research Gap

- **High Computational Burden of State-of-the-Art Detectors:** Existing high-accuracy object detection models (e.g., standard YOLOv3 with Darknet-53 or two-stage detectors like Faster R-CNN) are too resource-intensive and computationally heavy to run effectively in real-time on mobile or embedded devices used by visually impaired individuals.
- **Limitations of Prior Machine Learning Methods:** Traditional indoor navigation systems struggle to generalize across complex interior environments, suffering from poor detection performance under changing illumination, varied object textures/materials, heavy occlusions, and overlapping objects.
- **Inadequate Coverage of Essential Assistive Classes:** Existing public indoor datasets and state-of-the-art frameworks lack coverage for many specific, high-value landmark object classes and dangerous scenarios (e.g., podotactile tape, confidence zones, security buttons, disabled exits) vital for everyday indoor mobility for VIPs.

Research Questions

- **1:** How effectively can replacing Darknet-53 with a lightweight MobileNet V1 backbone in a YOLOv3 architecture reduce computational overhead while preserving high detection accuracy in indoor settings?
- **2:** To what extent does integrating Feature Pyramid Network (FPN)-like multi-scale feature representation improve the recognition of indoor objects varying in size, resolution, illumination, and occlusion states?
- **3:** Can an optimized, end-to-end deep learning framework trained on a specialized 25-class dataset achieve real-time execution speeds (FPS suitable for mobile deployment) and superior detection accuracy compared to existing baseline models?

My questions

3

Title: An End-to-End Object Detection System in Indoor Environments Using Lightweight Neural Network

Main Goal: To design and deploy a lightweight, efficient deep learning system along with a comprehensive indoor dataset to enhance real-time indoor navigation, independence, and safety for visually impaired and blind individuals (as well as sighted users).

Project Objective

- **Lightweight Model Architecture:** Re-engineer the YOLOv3 object detection framework by swapping its heavy Darknet-53 backbone with MobileNet V1 to reduce parameter count and computational complexity, making it suitable for resource-constrained devices.
- **Multi-Scale Real-Time Detection:** Utilize depthwise separable convolutions and a Feature Pyramid Network (FPN)-style architecture to enable fast inference speeds (up to 59 FPS) and high precision across small, medium, and large objects.
- **New Dataset Development:** Create and annotate a specialized dataset of 11,000 images covering 25 crucial indoor landmark classes to train and evaluate deep neural networks.

Research Gap

- **Lack of Essential Class Coverage:** Existing indoor object detection models rarely cover the full spectrum of specific indoor landmarks and hazardous obstacles (e.g., podotactile tape, confidence zones, security buttons, disabled exits) critical for visually impaired mobility.
- **Poor Generalization in Complex Indoor Settings:** Conventional machine learning methods and existing deep learning systems struggle with varying lighting conditions, severe occlusions, overlapping objects, and inter/intra-class variations.
- **Computational Overhead on Mobile Devices:** High-accuracy detectors (such as standard YOLOv3 with Darknet-53 or two-stage detectors) are computationally heavy and resource-intensive, making them ill-suited for low-latency, real-time deployment on mobile/embedded assistive hardware.

Research Questions

1. How effectively can replacing Darknet-53 with MobileNet V1 in the YOLOv3 architecture reduce computational demands while retaining high detection accuracy for indoor objects?
2. How can an FPN-like structure improve multi-scale detection performance under challenging real-world conditions such as low/varying light, heavy occlusion, and changing viewpoints?
3. Can the proposed lightweight framework deliver real-time inference speeds (sufficient FPS) and high mean Average Precision (mAP) across 25 indoor landmark categories to reliably assist target users during navigation?